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Finding errors in code

D0-L1 Bugs, errors and failures, the words and the method

Reproduce small, read the evidence, test one hypothesis at a time, pin the bug with a failing test, fix the cause, and never silence an error with a default.

Three kinds of problem, cheapest first	Syntax error (nothing runs), exception (crash with a traceback), wrong result (silent bug).
Defect, failure, root cause	Defect: the mistake in code. Failure: what someone sees. Root cause: the first place it went wrong.
The debugging method	Reproduce small, read the evidence, one hypothesis, one check, a failing test, fix the cause, look for the same mistake elsewhere.
Why except Exception: return 0.0 is dangerous	It turns a loud, specific failure into a silent wrong result.

D0-L2 Read code you did not write

Read the contract, the inputs and outputs and the side effects first, trace a tiny input by hand, then probe the edges and list the assumptions.

Order for reading unfamiliar code	Contract, inputs and outputs, side effects, a hand trace on a tiny input, edges, assumptions. Then run it on a known answer.
Side effects to look for	Writes to files or tables, network calls, prints, and changes to arguments, such as list.remove or df[col] = ... on a frame passed in.
Why not remove items while looping over a list	Each removal shifts the rest left, so the loop skips the next element.
How to trace by hand	Pick three values and write each variable's value in a table after every step.